Data Science Home Assignment

Overview

We want this assignment to give you a small taste of the kind of problems we work on in our team. If you proceed to the next stage, we will discuss your solution in detail in the following interview.

Tufin helps organizations define and manage security zones and segmentation policies across their network environments. These zones can be used for monitoring, policy control, and compliance use cases.

In practice, many clients find it difficult to segment their network space meaningfully and consistently at scale. This assignment is inspired by that type of problem: using observed security policy data to help infer suitable security-zone assignments for subnets.

In real environments, the data is large, noisy, and only partially labeled. For this exercise, we are giving you a small, simplified dataset of security rules together with partial tags. A rule represents an allowed connection between a source entity and a destination entity.

Your goal is to analyze the data, implement a reasonable solution for predicting tags for unlabeled entities in the dataset, and evaluate the quality and limitations of the results.

To keep the exercise focused, please assume that:

- all rules are allow rules
- all rules use the same service: HTTPS (TCP/443)

We would like to see how you think, how you approach imperfect data, how you evaluate results, and how you reason about limitations and trade-offs.

Please aim to spend **no more than 5 hours** on this assignment.

We do not expect a complete or polished solution.

We care more about your reasoning, judgment, and ability to work with imperfect data than about building the most complex model.

If you do not have time to finish everything, please clearly describe:

- what you completed
- what you did not complete
- what you would do next with more time

The Dataset

You are given two files:

1. rules.csv- contains allowed connections between objects:
   - rule_id
   - src_obj_id
   - dst_obj_id
2. tags.csv- contains known tags for part of the objects:
   - obj_id
   - tag

The Task

Please build and implement reasonable approach for assigning a tag to each unlabeled object.

Your solution may use any approach you think is appropriate.

Examples include supervised learning, clustering, heuristics, graph-based reasoning, or a combination of methods.

What we would like to see

1. **EDA** - We are not looking for a long exploratory report, only the main points.

Highlight anything important you notice about:

- the labeled data
- the unlabeled data
- the structure of the rules
- any limitations, risks, or suspicious patterns

2. **Method** - Describe the approach you chose and why.

We are interested in:

- how you framed the problem
- why you chose this method
- what alternatives you considered, if any

3. **Predictions** - produce predictions for the unlabeled objects.

Please provide a CSV named predictions.csv with the following columns:

- obj_id
- predicted_tag

4. **Evaluation** - evaluate your approach and explain the results. Please describe:
    - how you evaluated the method
    - which metrics you used
    - what the results mean
    - what they do not mean
    - any assumptions or caveats that affect how much the results can be trusted

Please also provide a CSV named metrics.csv with the following columns:

- metric
- value

This file should include the main metrics you used to evaluate your approach.

5. **Discussion** - briefly discuss:
    - key limitations of your solution
    - important data issues or risks
    - what you would improve or investigate next with more time

<u>Deliverables</u>

Please submit:

- code (notebook or Python scripts are both fine)
- a short write-up
- predictions.csv
- metrics.csv

** *<u>Notes</u>*

- You may use any Python libraries you find appropriate.
- We do not expect production-grade code.
- A simple, well-reasoned solution is better than a complex solution that is not well explained.
- If you choose to simplify the problem, that is fine, as long as you explain your reasoning.